Robust and Uncertainty-Aware VAE (RU-VAE) for One-Class Classification

Renuka Sharma[1,2] and Suyash P. Awate[1]

[1]Computer Science and Engineering Department, Indian Institute of Technology (IIT) Bombay, Mumbai.

[2]IITB-Monash Research Academy, Mumbai.

Introduction

- We propose a novel semi-supervised variational learning DNN framework leveraging
  - generalized-Gaussian (GG) models on encodings in the latent space and the reconstructed image, and
  - a classifier in the latent space.
- We propose a variational autoencoder (VAE) framework, RU-VAE that leads to data-adaptive modelling by subsuming:
  - robust modelling through the GG's shape parameters and
  - uncertainty-aware modelling through the GG's scale parameters.
- In RU-VAE, we have a classifier in the latent-space to discriminate between the embeddings of the inlier and outlier classes, thus refining the enclosing boundary for the normal data points.
- We leverage some outlier data during training to improve performance.
- To enable backpropagation-based optimization, we propose a reparameterization of the GG.
- We show improvements on the real-world medical datasets: Malaria and Nanofibre.

Proposed: Robust and Uncertainty-Aware VAE (RU-VAE)

- S: Set of inlier data points
- T: Set of outlier data points
- N: Number of inlier data points
- M: Number of outlier data points
- $\theta^{\Phi\Psi}$: Parameters of the generalized-Gaussian (GG) distribution output by the autoencoder RU-VAE
- θ^{Φ}: Parameters of the generalized−Gaussian (GG) distribution output by Encoder
- θ^{Ψ}: Parameters of the generalized-Gaussian (GG) distribution output by the Decoder
- $\Phi^{\mu}, \Phi^{\alpha}, \Phi^{\beta}$: Mean, scale, and shape parameters of the Encoder for RU-VAE
- $\Psi^{\mu}, \Psi^{\alpha}, \Psi^{\beta}$: Mean, scale, and shape parameters of the Decoder for RU-VAE
- $G(.;\mu,\alpha,\beta)$: Factorized generalized-Gaussian with mean vector μ, a vector of log-scale parameters α, and a vector of log-shape parameters β
- z_{ni}: Sampled inlier data point
  For each S_n, z_{ni} is the i-th independent draw from the probability density function $Q(Z|S_n;\theta^{\Phi})$
- I: Number of samples taken for inlier data points
- z_{mj}: Sampled outlier data point
  For each T_m, z_{mj} is the j-th independent draw from the probability density function $Q(Z|T_m;\theta^{\Phi})$
- J: Number of samples taken for outlier data points

RU-VAE is our proposed solution for semi-supervised anomaly detection where we have training data from the inlier class (S) and the outlier class (T). We minimize the loss for the normal data points (S) using the encoder-decoder based network using GG modeling and variational learning:

$$\mathcal{L}_1(\{S_n\}_{n=1}^N;\theta^{\Phi\Psi}) := \frac{1}{NI}\sum_{n=1}^{N}\sum_{i=1}^{I}[0.5\|z_{ni}(S_n,\theta^{\Phi})\|_2^2 + \log \mathcal{G}(z_{ni}(S_n,\theta^{\Phi});\Phi^{\mu}(S_n;\theta^{\Phi}),\Phi^{\alpha}(S_n;\theta^{\Phi}),\Phi^{\beta}(S_n;\theta^{\Phi})) - \log \mathcal{G}(S_n;\Psi^{\mu}(Z_n;\theta^{\Psi}),\Psi^{\alpha}(Z_n;\theta^{\Psi}),\Psi^{\beta}(Z_n;\theta^{\Psi}))]$$

Similar to $\mathcal{L}_1()$, we propose a loss term for the expert labeled outliers' set as $\mathcal{L}_2()$. Because we want the outliers to be away from the compact subspace containing the inliers, we consider the negation of the terms (compared to $\mathcal{L}_1()$) for latent-space and image-space for the outliers as follows.

$$\mathcal{L}_2(\{T_m\}_{m=1}^M;\theta^{\Phi\Psi}) := -\frac{1}{MJ}\sum_{m=1}^{M}\sum_{j=1}^{J}[0.5\|z_{mj}(T_m,\theta^{\Phi})\|_2^2 + \log \mathcal{G}(z_{mj}(T_m,\theta^{\Phi});\Phi^{\mu}(T_m;\theta^{\Phi}),\Phi^{\alpha}(T_m;\theta^{\Phi}),\Phi^{\beta}(T_m;\theta^{\Phi})) - \log \mathcal{G}(T_m;\Psi^{\mu}(Z_m;\theta^{\Psi}),\Psi^{\alpha}(Z_m;\theta^{\Psi}),\Psi^{\beta}(Z_m;\theta^{\Psi}))]$$

To promote separability of the normal-class and abnormal class distributions in latent space, we propose another loss term. We introduce a nonlinear classifier $C(.;\theta^{C})$ to discriminate between the distributions of the latent-space encodings of the independent draws. The associated cross-entropy is as follows.

$$\mathcal{L}_3(\{S_n\}_{n=1}^N,\{T_m\}_{m=1}^M;\theta^{\Phi\Psi}) := -\frac{1}{MJ}\sum_{m=1}^{M}\sum_{j=1}^{J}\log(\mathcal{C}(z_{mj}(T_m,\theta^{\Phi});\theta^{\mathcal{C}})) - \frac{1}{NJ}\sum_{n=1}^{N}\sum_{j=1}^{J}\log(1-\mathcal{C}(z_{ni}(S_n,\theta^{\Phi});\theta^{\mathcal{C}}))$$

Thus, RU-VAE learning minimizes

$$\lambda_1\mathcal{L}_1(\{S_n\}_{n=1}^N;\theta^{\Phi\Psi}) + \lambda_2\mathcal{L}_2(\{T_m\}_{m=1}^M;\theta^{\Phi\Psi}) + \mathcal{L}_3(\{S_n\}_{n=1}^N,\{T_m\}_{m=1}^M;\theta^{\Phi\Psi}),$$

where $\lambda_1 > 0$ and $\lambda_2 > 0$ are the weighing factors for the objective function above.
RU-VAE leverages factored GG models for, both, robust heavy-tailed modeling (through GG shape parameters) and uncertainty-aware heteroscedastic modeling (through GG scale parameters) in latent space and image space; the GG models adapt to the specific input.

Thanks to the Infrastructure Facility for Advanced Research and Education in Diagnostics grant funded by Department of Biotechnology, Government of India (BT/INF/22/SP23026/2017).

Proposed Learning Framework

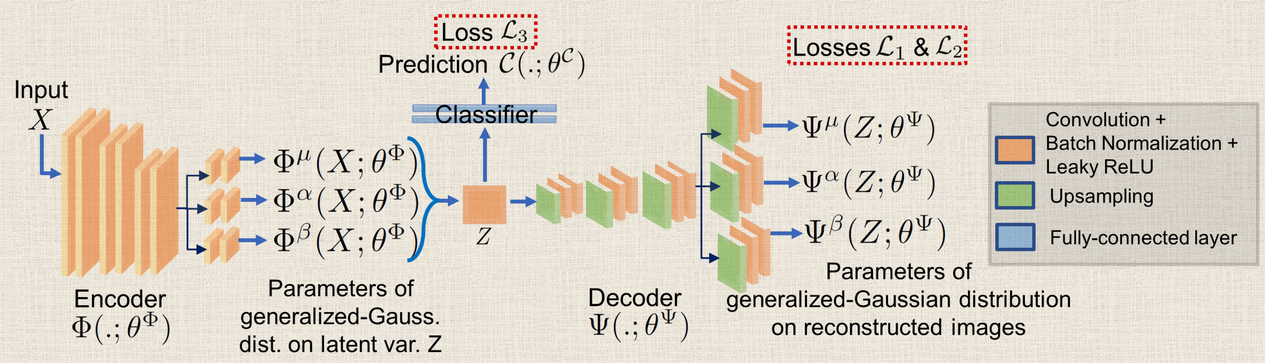


Fig: Robust and Uncertainty-aware VAE (RU-VAE) Framework for OCC, a semi-supervised anomaly detection based approach. The input X can be inlier or outlier and we have losses in the latent-space as well as the image-space. Given a test image U, we classify it as inlier/outlier based on an anomaly score.

Reparameterization Scheme

- We reparameterize z_{ni} based on Gamma random variables as

  $[z_{ni}]_d := [\Phi^{\mu}(X_n;\theta^{\Phi})]_d + (\Delta + [\exp(\Phi^{\alpha}(X_n;\theta^{\Phi}))]_d)\, W_{nid}\, |Y_{nid}|^{1/(\tau+\exp([\Phi^{\beta}(X_n;\theta^{\Phi})]_d))}$,

  where random variables Y_{nid} have the PDF $\mathrm{Gamma}(1/(\tau + [\Phi^{\beta}(S;\theta^{\Phi})]_d), 1)$ and W_{nid} are Bernoulli random variables taking a value 1 with probability 0.5.
- We leverage implicit reparameterization gradients for the Gamma random variables Y_{nid}.

Experimental setup

We are using two medical datasets to show the significance of our method for AD:

- **Malaria dataset:** The Broad Bioimage Benchmark Collection has 1328 microscopy images of blood, each containing red blood cells manually segmented and labelled as malaria-infected (abnormal) or not (normal); we classify patches of size 170x170 pixels (cell diameter is typically 170 pixels); we randomly extract 10000 normal patches and 2500 abnormal patches. Infected RBCs are of 6 types, with varying natural frequency of occurrence. To realize a OCC scenario, training uses abnormal patches from only 3 infected-RBC types, and testing uses patches from all 6 infected-RBC types.
- **Nanofibre dataset: T**he Nanofiber has 45 electron-microscopy images of nanofiber biomaterials, with 40 images showing material defects (manually segmented); analogous to the malaria dataset, we randomly extract 17228 normal patches and 5452 abnormal patches of size 128x128 pixels. All methods resize patches to 32x32 pixels. We partition the datasets into 3 subsets for training, validation (to tune free parameters), and testing (test set contains an equal number of normal and abnormal patches).

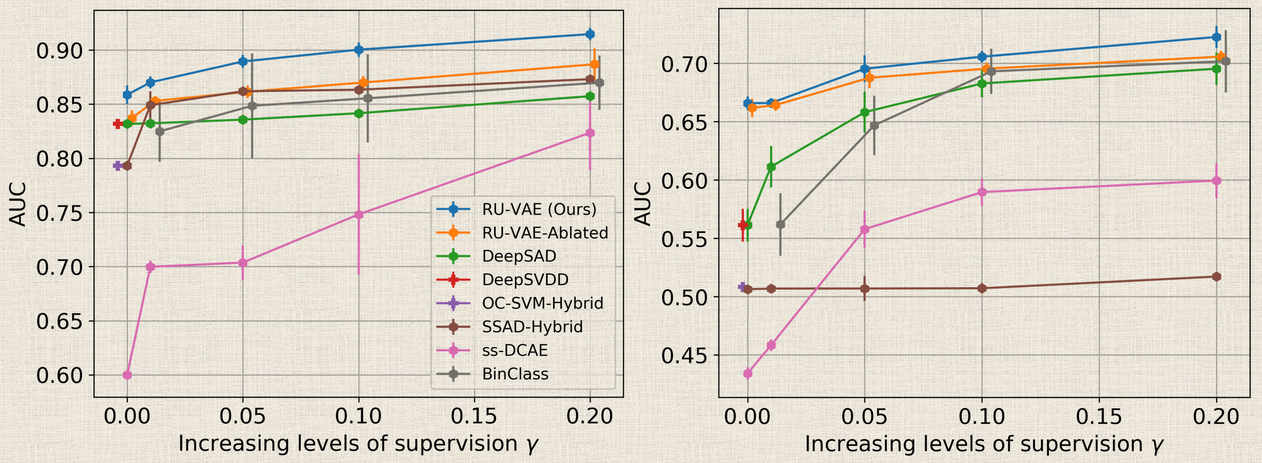
Inference Strategy

For a test image U, we propose an abnormality score as the norm of the mean $\Phi^{\mu}(U;\theta^{\Phi})$ (outliers lead to larger scores as compared to inliers), and then classify based on a threshold (free parameter) on the score.

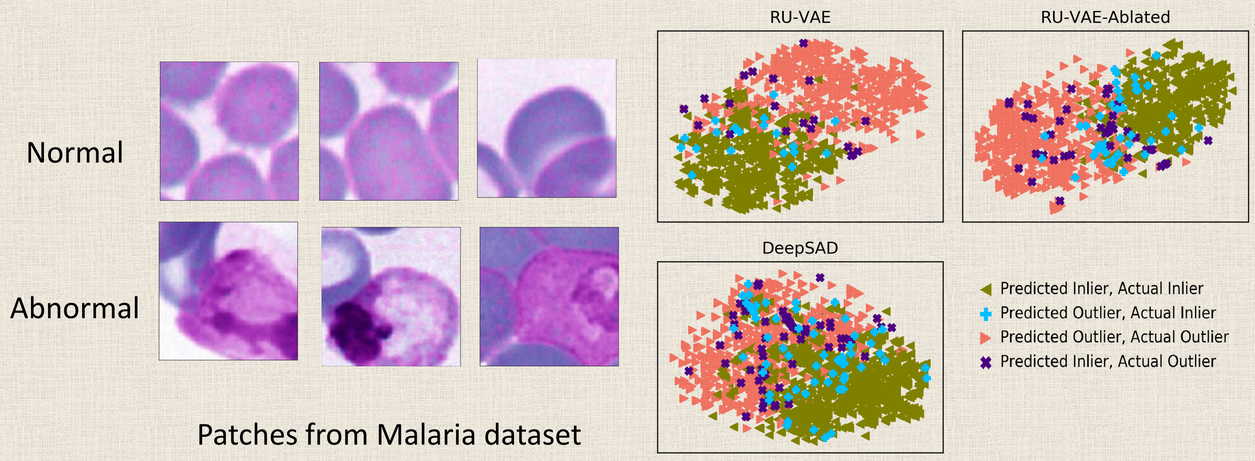
Optimization Strategy

Our optimization scheme is hierarchical. It first uses solely the inlier training set and excludes the branches modeling log-scale and log-shape (fixing GG scale parameters to 1 and shape parameters to 2). With this warm start, we include the log-scale and log-shape branches and retrain the DNN. Last, we include the outlier training set and the classifier branch and retrain the DNN.

Quantitative results on Malaria and Nanofibre dataset



Qualitative results on Malaria dataset



Patches from Malaria dataset

Conclusive Remarks

Comprehensive empirical analyses, involving 6 baselines and an ablated version on two public datasets, show the benefits of our method over the state of the art and provide insights into the benefits of the novel components.